Stroma World — Module Edge List v1.0

> **What this is:** Every directed causal connection between the 8 biological modules. **Format:** Source → Target, what the edge carries, whether it increases or decreases the target's output, and confidence that this interaction matters in PDAC stroma biology. **Rule:** If an edge isn't listed here, those two modules don't talk. No implicit connections.

Edge Anatomy

Field	Meaning
Edge ID	Unique reference (E01, E02, ...)
Source → Target	Which module sends, which receives
Carrier	What travels along the edge (a signal, a state, a field value)
Effect	INCREASES or DECREASES the target module's primary output
Conditionality	Always active, or only under specific conditions
Confidence	HIGH / MED / LOW that this edge is real and significant in PDAC
Anchors	Which Reality Anchors depend on this edge being correct

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The Edge List

Module 1 (O_2 Sensing) → Others

ID	Source	Target	Carrier	Effect on Target	Conditionality	Confidence	Anchors
E01	M1 (O_2 Sensing)	M5 (EMT Engine)	HIF1A=ON state	**INCREASES** EMT induction signal	Only when O_2 below threshold	HIGH	5, 6
E02	M1 (O_2 Sensing)	M2 (Paracrine Secretion)	HIF1A=ON state	**INCREASES** TGF-β secretion rate in tumor cells	Only when O_2 below threshold	HIGH	1, 5
E03	M1 (O_2 Sensing)	M4 (Prolif & Death)	HIF1A=ON state	**DECREASES** proliferation rate (acute hypoxia slows cycling)	Only when O_2 below threshold	MED	5
E04	M1 (O_2 Sensing)	M4 (Prolif & Death)	HIF1A=ON state	**INCREASES** death resistance (survival adaptation)	Only when O_2 below threshold	HIGH	5, 10
E05	M1 (O_2 Sensing)	M7 (Drug Response)	HIF1A=ON state	**INCREASES** NRF2 priming speed (hypoxia pre-arms stress response)	Only when O_2 below threshold AND drug present	MED	10

Note on E03 vs. E04: These are not contradictory. Acute hypoxia simultaneously slows cycling (E03) and improves survival (E04). The net effect is a cell that divides less but dies less — a quiescent, resistant

phenotype. Both edges are necessary to reproduce the hypoxic sanctuary behavior in Anchor 10.

Module 2 (Paracrine Secretion) → Others

ID	Source	Target	Carrier	Effect on Target	Conditionality	Confidence	Anchors
E06	M2 (Paracrine Secretion)	M3 (Stromal Activation)	TGF-β in microenvironment field	**INCREASES** probability of PSC → CAF transition	Always (tumor constitutively secretes)	HIGH	1, 4, 9
E07	M2 (Paracrine Secretion)	M3 (Stromal Activation)	SHH in microenvironment field	**INCREASES** probability of PSC → CAF transition and GLI1 activation	Always (tumor constitutively secretes)	HIGH	1, 3
E08	M2 (Paracrine Secretion)	M5 (EMT Engine)	TGF-β in microenvironment field (SMAD-independent arm)	**INCREASES** EMT induction signal in tumor cells	Always, but EMT effect is via SMAD-independent pathway; works regardless of SMAD4 status	HIGH	6

ID	Source	Target	Carrier	Effect on Target	Conditionality	Confidence	Anchors
E09	M2 (Paracrine Secretion)	M4 (Prolif & Death)	TGF-β in microenvironment field (SMAD-dependent arm)	**DECREASES** tumor proliferation rate (growth arrest)	ONLY when SMAD4 = WT in receiving cell; completely absent when SMAD4 = LOF	HIGH	4

Critical note on E08 vs. E09: These are two edges from the same source (TGF-β) to different targets, carrying different arms of TGF-β signaling. E09 (brake) is conditional on SMAD4. E08 (EMT) is unconditional. In canonical PDAC (SMAD4=LOF), E09 is severed and E08 remains — this is the SMAD4 asymmetry. Both edges must exist in the wiring, and E09 must be gated by SMAD4 status. Getting this wrong breaks Anchor 4.

Module 3 (Stromal Activation) → Others

ID	Source	Target	Carrier	Effect on Target	Conditionality	Confidence	Anchors
E10	M3 (Stromal Activation)	M6 (ECM Production)	ACTA2=ON state	**INCREASES** ECM production rate (both HA and collagen)	Only in activated CAFs	HIGH	1, 2, 7, 8, 9
E11	M3 (Stromal	M6 (ECM Production)	GLI1=ON state	**INCREASES** ECM	Only when SHH pathway	HIGH	3, 8

ID	Source	Target	Carrier	Effect on Target	Conditionality	Confidence	Anchors
	Activation)			production rate (additive boost on top of ACTA2)	is active in activated CAFs		
E12	M3 (Stromal Activation)	M2 (Paracrine Secretion)	CAF identity (ACTA2=ON)	**INCREASES** TGF-β secretion from stromal cells (positive feedback)	Only in activated CAFs	HIGH	1, 7
E13	M3 (Stromal Activation)	M4 (Prolif & Death)	CAF identity + GLI1 state	**INCREASES** stromal cell proliferation rate (more CAFs = more barrier)	Only in activated CAFs; boosted if GLI1=ON	MED	1, 3

E12 is the most dangerous edge in the system. It creates the TGF-β positive feedback loop: tumor secretes TGF-β → activates CAFs → CAFs secrete more TGF-β → activates more CAFs → ... Without E12, the barrier reaches a limited size and stops. With E12, the barrier self-amplifies. This edge is why PDAC stroma is uniquely dense.

Module 4 (Prolif & Death) → Others

ID	Source	Target	Carrier	Effect on Target	Conditionality	Confidence	Anchors
E14	M4	M8	Cell	**INCREASES**	Only when cells	HIGH	3

ID	Source	Target	Carrier	Effect on Target	Conditionality	Confidence	Anchors
	(Prolif & Death)	(Mechanical Compaction)	division events (tumor mass increase)	solid stress at tumor-stroma boundary	are actively dividing in a confined space		
E15	M4 (Prolif & Death)	M2 (Paracrine Secretion)	Cell survival (alive cell continues secreting)	**INCREASES** total paracrine output (more living cells = more signal)	Always (every living tumor cell secretes)	HIGH	1, 10
E16	M4 (Prolif & Death)	M2 (Paracrine Secretion)	Cell death (dead cell stops secreting)	**DECREASES** total paracrine output (fewer living cells = less signal)	Only when cells die (apoptosis from drug or stochastic death)	HIGH	7, 10

E15 and E16 are the life/death balance of the barrier. If the tumor dies too fast (E16 dominates), paracrine drive collapses and the stroma resolves. If the tumor survives (E15 dominates), secretion continues and the barrier is sustained. This is why baseline apoptotic resistance (Knob 4) matters for barrier maintenance, not just for tumor survival per se.

Module 5 (EMT Engine) → Others

ID	Source	Target	Carrier	Effect on Target	Conditionality	Confidence	Anchors
E17	M5 (EMT Engine)	M4 (Prolif & Death)	ZEB1=ON state	**DECREASES** proliferation rate	Only in mesenchymal (ZEB1=ON) cells	MED	6

ID	Source	Target	Carrier	Effect on Target	Conditionality	Confidence	Anchors
				(go-vs-grow tradeoff)			
E18	M5 (EMT Engine)	M4 (Prolif & Death)	ZEB1=ON state	**INCREASES** death resistance (anoikis resistance)	Only in mesenchymal cells	MED	6, 10
E19	M5 (EMT Engine)	M6 (ECM Production)	MMP2 activity flag	**DECREASES** local ECM density (matrix degradation)	Only in mesenchymal cells with MMP2 active (extent = MEDIUM or HIGH)	HIGH	6, 7

E17 and E18 parallel E03 and E04 (the hypoxia pair). EMT, like hypoxia, produces cells that divide less but survive more. The invasive front is populated by slow-cycling, hard-to-kill, motile cells — not by the fast-proliferating cells in the tumor core. This spatial division of labor is what makes PDAC so difficult: the front that meets therapy or immune cells is armored.

Module 6 (ECM Production) → Others

ID	Source	Target	Carrier	Effect on Target	Conditionality	Confidence	Anc
E20	M6 (ECM Production)	M1 (O_2 Sensing)	ECM density increase → O_2 diffusion coefficient decrease	**INCREASES** likelihood of HIF1A activation (less O_2	Always proportional to ECM density	HIGH	2, 5

ID	Source	Target	Carrier	Effect on Target	Conditionality	Confidence	Anc
				reaches tumor interior)			
E21	M6 (ECM Production)	M7 (Drug Response)	ECM density increase → Drug diffusion coefficient decrease	**DECREASES** drug concentration at tumor cells behind barrier	Always proportional to ECM density	HIGH	2, 9,
E22	M6 (ECM Production)	M8 (Mechanical Compaction)	ECM stiffness (collagen component)	**INCREASES** mechanical resistance to tumor expansion	Always proportional to collagen fraction of ECM	HIGH	3, 8
E23	M6 (ECM Production)	M5 (EMT Engine)	ECM density at tumor-stroma interface → TGF-β trapping/concentration	**INCREASES** local TGF-β concentration near dense ECM (ECM sequesters and concentrates TGF-β)	Always; effect is proportional to ECM density	MED	6

E20 is the hypoxia amplification trigger. More ECM → less O_2 diffusion → more hypoxia → more HIF1A → more TGF-β secretion (E02) → more stroma activation (E06) → more ECM (E10) → even less O_2. This is the positive feedback loop that makes the barrier self-reinforcing. E20 is its entry point.

E23 is a subtle but important edge. Dense ECM doesn't just block diffusion — it also *concentrates* locally secreted TGF-β by slowing its diffusion away from the source. This means the tumor-stroma

interface has higher TGF-β concentration than the open space, which explains why EMT (E08) and stromal activation (E06) are strongest right at the boundary. Confidence is MED because the magnitude of this trapping effect is hard to measure, but the direction is well-established.

Module 7 (Drug Response) → Others

ID	Source	Target	Carrier	Effect on Target	Conditionality	Confidence	Anchors
E24	M7 (Drug Response)	M4 (Prolif & Death)	Intracellular drug kill signal	**INCREASES** apoptosis rate	Only when drug is present at sufficient intracellular concentration	HIGH	7, 9, 10
E25	M7 (Drug Response)	M4 (Prolif & Death)	ABCB1=ON state	**DECREASES** effective drug kill signal (efflux reduces intracellular drug)	Only when ABCB1 has been induced (delay elapsed, NRF2 active)	HIGH	10

E24 and E25 are the drug efficacy tug-of-war. E24 pushes toward cell death (drug works). E25 pushes toward survival (resistance kicks in). The temporal dynamics matter: E24 acts immediately when drug arrives, E25 kicks in after the induction delay (Knob 7a). The EA should discover that exploiting this delay window — pulsed dosing, cycling, or efflux inhibitors — is more effective than continuous exposure.

Module 8 (Mechanical Compaction) → Others

ID	Source	Target	Carrier	Effect on Target	Conditionality	Confidence	Anchors
E26	M8 (Mechanical Compaction)	M4 (Prolif & Death)	Solid stress exceeds contact inhibition threshold	**DECREASES** proliferation rate (growth arrest under compression)	Only when tumor cells are confined by stiff ECM	HIGH	3
E27	M8 (Mechanical Compaction)	M6 (ECM Production)	Compression of existing ECM	**INCREASES** local ECM density (passive densification, not new production)	Only at tumor-stroma boundary where expansion presses against barrier	MED	3, 8
E28	M8 (Mechanical Compaction)	M1 (O_2 Sensing)	Increased ECM density from compaction → further reduced O_2 diffusion	**INCREASES** likelihood of HIF1A activation (compaction-induced hypoxia)	Indirect: via E27 increasing ECM → E20 reducing O_2 diffusion	MED	5

E26 is why stroma restrains the tumor. Without this edge, removing stroma (Anchor 3, SHH inhibition) wouldn't cause tumor acceleration — there would be nothing holding the tumor back. E26 encodes the physical confinement that makes the vismodegib paradox possible. If Anchor 3 fails, check E26 first.

E28 is a second-order edge (compaction → ECM density → O_2 diffusion → hypoxia). It doesn't carry a novel signal — it amplifies E20 by adding a mechanical contribution to ECM density. Confidence is MED because the relative magnitude of compaction-driven vs. CAF-production-driven ECM densification is debated, but the direction is not.

Master Edge Table

ID	Source → Target	Effect	Conditionality	Confidence	Key Anchors
E01	M1 → M5	INCREASES EMT induction	Hypoxia	HIGH	5, 6
E02	M1 → M2	INCREASES TGF-β secretion	Hypoxia	HIGH	1, 5
E03	M1 → M4	DECREASES proliferation	Hypoxia	MED	5
E04	M1 → M4	INCREASES death resistance	Hypoxia	HIGH	5, 10
E05	M1 → M7	INCREASES NRF2 priming	Hypoxia + drug	MED	10
E06	M2 → M3	INCREASES stromal activation	Always	HIGH	1, 4, 9
E07	M2 → M3	INCREASES stromal activation (SHH arm)	Always	HIGH	1, 3
E08	M2 → M5	INCREASES EMT (SMAD-independent)	Always	HIGH	6
E09	M2 → M4	DECREASES proliferation (TGF-β brake)	SMAD4=WT only	HIGH	4
E10	M3 → M6	INCREASES ECM production	CAF activated	HIGH	1, 2, 7, 8, 9

ID	Source → Target	Effect	Conditionality	Confidence	Key Anchors
E11	M3 → M6	INCREASES ECM production (GLI1 boost)	SHH pathway active	HIGH	3, 8
E12	M3 → M2	INCREASES TGF-β secretion (CAF feedback)	CAF activated	HIGH	1, 7
E13	M3 → M4	INCREASES stromal proliferation	CAF activated	MED	1, 3
E14	M4 → M8	INCREASES solid stress	Cells dividing in confinement	HIGH	3
E15	M4 → M2	INCREASES paracrine output (cell alive)	Always	HIGH	1, 10
E16	M4 → M2	DECREASES paracrine output (cell dead)	Cell death event	HIGH	7, 10
E17	M5 → M4	DECREASES proliferation (go-vs-grow)	ZEB1=ON	MED	6
E18	M5 → M4	INCREASES death resistance (anoikis-R)	ZEB1=ON	MED	6, 10
E19	M5 → M6	DECREASES local ECM (MMP2)	EMT extent ≥ MEDIUM	HIGH	6, 7
E20	M6 → M1	INCREASES hypoxia likelihood	Always ∝ ECM density	HIGH	2, 5
E21	M6 → M7	DECREASES drug at tumor cells	Always ∝ ECM density	HIGH	2, 9, 10

ID	Source → Target	Effect	Conditionality	Confidence	Key Anchors
E22	M6 → M8	INCREASES mechanical resistance	Always ∝ collagen fraction	HIGH	3, 8
E23	M6 → M5	INCREASES local TGF-β concentration (trapping)	Always ∝ ECM density	MED	6
E24	M7 → M4	INCREASES apoptosis (drug kill)	Drug present	HIGH	7, 9, 10
E25	M7 → M4	DECREASES drug kill (efflux)	ABCB1 induced	HIGH	10
E26	M8 → M4	DECREASES proliferation (contact inhibition)	High solid stress	HIGH	3
E27	M8 → M6	INCREASES ECM density (passive compaction)	Tumor expanding against stiff ECM	MED	3, 8
E28	M8 → M1	INCREASES hypoxia (via compaction-driven ECM)	Indirect via E27 → E20	MED	5

Totals: 28 directed edges. 17 INCREASES. 11 DECREASES. 19 HIGH confidence. 9 MED confidence. 0 LOW confidence.

Confidence Breakdown

HIGH Confidence (19 edges)

These are well-established interactions in PDAC biology with strong experimental support from multiple independent studies. If the simulation omits any of these, it is missing known biology.

MED Confidence (9 edges)

These are directionally established but with less certainty about their relative importance in PDAC specifically. Each one is flagged below with a note on what could change it to HIGH or cause it to be removed:

Edge	Why MED	What would promote to HIGH	What would justify removal
E03	Acute hypoxia growth slowing is well-known but magnitude in PDAC context is debated; some PDAC cells adapt quickly and maintain proliferation	Time-course proliferation data under hypoxia in AsPC-1 / PANC-1	If PDAC cells show no measurable proliferation slowdown under acute hypoxia
E05	HIF1A → NRF2 cross-priming is established in pan-cancer but not deeply characterized in PDAC specifically	NRF2 activation kinetics under hypoxia in PDAC lines	If NRF2 activation is entirely drug-dependent with no hypoxia contribution
E13	CAF proliferation rate and its dependence on GLI1 is established but less studied than CAF ECM production	GLI1 knockdown effect on CAF doubling time in co-culture	If CAF proliferation proves negligible compared to CAF activation from quiescent PSC pool
E17	Go-vs-grow tradeoff is well-documented in glioma and breast cancer; less directly measured in PDAC	Proliferation rate measurement in ZEB1-high vs. ZEB1-low PDAC cells	If PDAC EMT cells show no proliferation penalty (some aggressive lines may not)
E18	Anoikis resistance from EMT is well-established pan-cancer; PDAC-specific magnitude is less characterized	Anoikis assay comparing epithelial vs. mesenchymal PDAC sublines	If PDAC cells are already anoikis-resistant at baseline (due to TP53/BCL-XL) making EMT contribution negligible
E23	ECM concentrating TGF-β by slowing diffusion is physically sound but hard to measure directly	TGF-β concentration profiling across ECM density gradients in tissue	If TGF-β diffusion is fast enough that trapping effect is

Edge	Why MED	What would promote to HIGH	What would justify removal
			negligible at PDAC ECM densities
E27	Passive mechanical compaction of ECM is physically real but its contribution relative to active CAF production is debated	Comparison of ECM density at tumor boundary with and without CAFs (mechanical-only vs. combined)	If mechanical contribution to ECM density is <5% of total (dominated by active production)
E28	Second-order edge (compaction → ECM → hypoxia); real but the unique contribution beyond E20 is uncertain	Oxygen measurements in tumors with high vs. low solid stress, controlling for CAF activity	If compaction-driven ECM densification proves negligible (E27 removed), E28 also falls

No LOW Confidence Edges

Every edge in the list has at least moderate evidence. Speculative interactions were excluded during module design. If an edge were LOW confidence, it wouldn't be in v1.0 — it would be flagged for investigation in a later cycle.

Feedback Loop Tracing

Each feedback loop in the system is composed of specific edges:

Loop 1: Stroma-Building Positive Feedback

E15 (living tumor cells) → E06 (TGF-β activates PSCs) → E10 (CAFs produce ECM) →
E12 (CAFs secrete more TGF-β) → E06 (more PSCs activated) → ...

All edges HIGH confidence. This loop is rock-solid — it's the most established feature of PDAC stroma biology.

Loop 2: Hypoxia Amplification Positive Feedback

E10 (CAFs produce ECM) → E20 (ECM blocks O_2) → E02 (HIF1A boosts TGF-β secretion) →
E06 (more stromal activation) → E10 (more ECM) → ...

All edges HIGH confidence. Second-strongest loop in the system.

Loop 3: Mechanical Confinement Negative Feedback

E14 (tumor growth generates stress) → E26 (stress inhibits proliferation) →
E14 reduced (less growth → less stress) → ...

Both edges HIGH confidence. Classic density-dependent growth regulation.

Loop 4: EMT Self-Limiting Negative Feedback

E01 (hypoxia → EMT) → E19 (EMT cells degrade local ECM via MMP2) →
E20 reversed locally (less ECM → more O_2 locally) → E01 reduced (less hypoxia → less EMT) → ...

E01 HIGH, E19 HIGH, E20 HIGH. Loop is high-confidence but self-limiting nature depends on MMP2 being locally effective — if MMP2 degrades too little ECM, this loop won't close and EMT will be persistent rather than reversible.

Loop 5: Drug Resistance Adaptive Loop

E21 (ECM blocks drug) → E24 attenuated (less drug reaches cell) →
E25 activated in surviving cells (ABCB1 induced) → E24 further attenuated →
E15 (surviving cell keeps secreting) → E06 (stroma maintained) → E21 (barrier persists) → ...

All edges HIGH confidence. This is the treatment failure cascade.

Edge Dependency on Reality Anchors

Which edges must be correct for each anchor to pass:

Anchor	Required edges (must be correct)	Edges that amplify but aren't strictly required
1 (Self-assembly)	E06, E07, E10, E12, E15	E02, E11, E13
2 (Drug diffusion drops)	E10, E20, E21	E27
3 (SHH paradox)	E07, E11, E14, E22, E26	E13, E27
4 (SMAD4 asymmetry)	E06, E08, E09	E10
5 (Central hypoxia)	E10, E20	E02, E27, E28
6 (Peripheral EMT)	E01, E08, E19, E23	E17, E18
7 (ECM degradation insufficient)	E10, E12, E19, E21, E24	E25
8 (HA vs. collagen)	E10, E11, E22	E27
9 (Barrier predicts failure)	E06, E10, E21, E24	E25
10 (Spatial sanctuary)	E04, E15, E21, E24, E25	E05, E12, E16, E18

E10 (CAF activation → ECM production) appears in 8 of 10 anchors. It is the single most critical edge in the system. If E10 is wrong, almost nothing works.

E06 (TGF-β → stromal activation) appears in 5 anchors. Second most critical.

E21 (ECM → drug diffusion block) appears in 4 anchors. Third most critical.

These three edges — E06, E10, E21 — are the load-bearing walls of the simulation. Validate them first, in that order.

Implementation Checklist

Priority 1 — Load-bearing edges (validate first):

- □ E06 TGF-β activates stroma
- □ E10 Activated CAFs produce ECM
- □ E21 ECM blocks drug diffusion

Priority 2 — Feedback loop closers:

- □ E12 CAF TGF-β feedback (closes Loop 1)
- □ E20 ECM blocks O_2 (closes Loop 2)
- □ E26 Solid stress inhibits proliferation (closes Loop 3)
- □ E19 MMP2 degrades local ECM (closes Loop 4)

Priority 3 — Signaling edges:

- □ E01 HIF1A → EMT
- □ E02 HIF1A → amplified secretion
- □ E07 SHH → stromal activation
- □ E08 TGF-β → EMT (SMAD-independent)
- □ E09 TGF-β → growth arrest (SMAD-dependent, conditional on SMAD4)

Priority 4 — Drug/resistance edges:

- □ E24 Drug kills cells
- □ E25 ABCB1 reduces drug kill
- □ E14 Proliferation generates solid stress
- □ E22 Collagen provides mechanical stiffness

Priority 5 — Amplifying/secondary edges:

- □ E03, E04, E05 Hypoxia secondary effects
- □ E11, E13 GLI1 boost effects
- □ E15, E16 Life/death → secretion balance
- □ E17, E18 EMT → proliferation/death tradeoff
- □ E23 ECM TGF-β trapping
- □ E27, E28 Mechanical compaction secondary effects

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28 edges. 17 increase, 11 decrease. 19 high confidence, 9 medium, 0 low. Five feedback loops. Three load-bearing edges. No edge exists without an anchor that needs it.